

THE GEORGE WASHINGTON UNIVERSITY
WASHINGTON, DC

Data Science Program
Department of Data Science
DATS 6501 – Capstone Project – Fall 2026

Post-Training Analysis & Novelty Detection Using Neural Network Self-Organizing Maps (NNSOM)

PHASE 3 REPORT

Novelty Detection via Quantization Error, U-Matrix and Baseline Evaluation

Capstone – Group 5

Nazish Atta
Fyrooz Khan

The George Washington University
Washington, DC

Supervised by
Dr. Amir Jafari
Associate Professor of Data Science

Draft for Issue #18; pending team and instructor review.

1. Overview and Scope

Following the project's Phase 3 objective, we examine whether a frozen self-organizing map (SOM) and comparison detectors can recognize feature representations that differ from MNIST. A trained MNIST convolutional classifier provides 84-dimensional `fc2` embeddings, and a frozen 15×15 SOM supplies the reference prototypes. MNIST training embeddings ($n = 49{,}000$) support fitting; MNIST validation ($n = 10{,}500$) forms the in-distribution (ID) evaluation set. Fashion-MNIST ($n = 10{,}000$) is the principal instructor-requested out-of-distribution (OOD) comparison. The proposal-specified SVHN ($n = 26{,}032$) is reported separately. These are development comparisons, not a final locked MNIST test result.

2. Related Work and Project Contribution

Kohonen introduced self-organizing maps, a topology-preserving organization of inputs into neuron prototypes (Kohonen 1990). Unlike work developing the SOM itself, this study applies an already-trained map as a frozen novelty reference. Vesanto and Alhoniemi investigated clustering of SOM representations (Vesanto and Alhoniemi 2000); our use of the U-Matrix focuses instead on local prototype separation as a candidate novelty signal.

LeCun and coauthors described convolutional approaches to document recognition (LeCun et al. 1998). In this project, an MNIST CNN is not trained for OOD classification; its frozen intermediate embeddings provide the shared space for downstream SOM and distance-based detectors. The NNSOM library documentation from Jafari provides the implementation context for the project's SOM component (Jafari n.d.); the contribution here is comparative post-training evaluation rather than a new SOM library.

Hendrycks and Gimpel introduced neural classification confidence as a simple OOD reference (Hendrycks and Gimpel 2017). We include max-softmax probability (MSP) as a baseline using the same frozen classifier, instead of treating SOM performance in isolation. Lee and coauthors proposed a Mahalanobis-based OOD score using neural representations (Lee et al. 2018); our comparison trains the relevant reference statistics on MNIST training embeddings. Liu and colleagues studied energy scores derived from classifier logits (W. Liu et al. 2020); we compare that complementary logit-based signal against prototype-distance and confidence-based scores. These literature methods motivate the baselines, but our measured comparisons are specific to the frozen MNIST checkpoint and evaluated datasets.

Isolation Forest uses random partitioning to isolate unusual observations (F. T. Liu, Ting, and Zhou 2008). Here we evaluate its novelty scores on the frozen CNN representations rather than raw features, and report its poor historical ranking without reversing the score after observing OOD labels. One-Class SVM estimates a support boundary for typical data (Schölkopf et al. 2001); our application fits that boundary on MNIST embeddings, while noting that its historical fitting budget differs from other baselines.

Sakurada and Yairi examined reconstruction-based anomaly detection using autoencoders (Sakurada and Yairi 2014). Our feature autoencoder similarly uses reconstruction error, but operates on frozen classifier embeddings and has an unresolved historical input-scaling limitation. Sun and colleagues applied deep nearest neighbors to OOD detection (Sun et al. 2022); we include a kNN-distance comparator in the same CNN feature space rather than claiming the SOM geometry is uniquely effective.

3. Methods and Frozen Experimental Setup

3.1 Representation, SOM and Quantization Error

The SOM has 225 units and uses an 84-dimensional representation. For scaled embedding $z(x)$ and SOM weights w_j, the best matching unit and quantization error are

$$b(x) = \arg\min_j \|z(x) - w_j\|_2, \qquad s_{\mathrm{QE}}(x) = \|z(x) - w_{b(x)}\|_2. \tag{1}$$

Higher QE means a sample's embedding lies farther from its nearest map prototype. The historical decision threshold is the 95th percentile of MNIST training QE, $\tau = 1.5690279077$; a sample with QE $> \tau$ is flagged. The instructor's later preference for choosing the threshold on validation instead of training differs from this frozen protocol, and that discrepancy is recorded in `src/v1_mnist/docs/DEVIATIONS.md`. The old threshold has not been retroactively tuned.

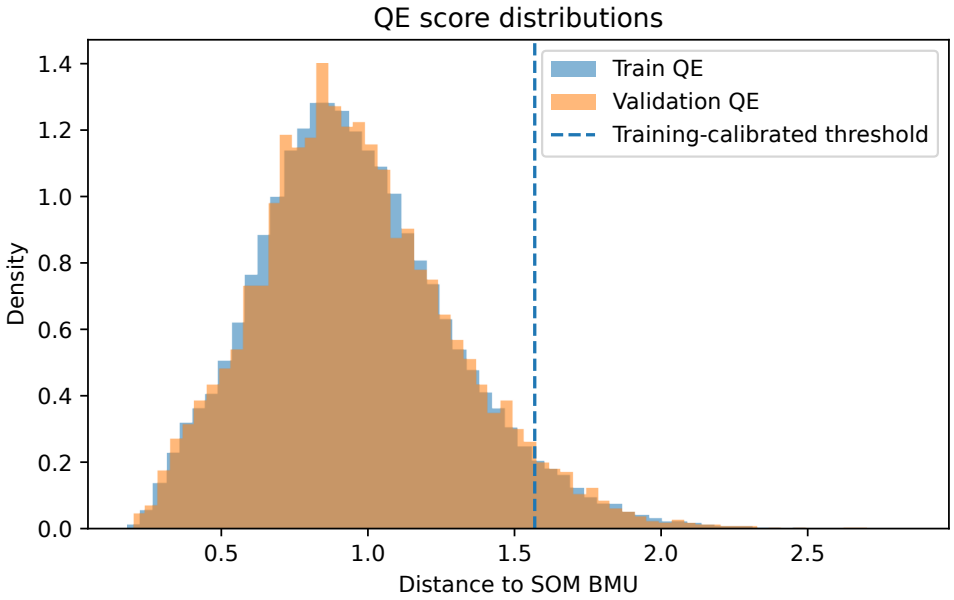


Figure 1. *Previously generated training and validation QE distributions with the frozen training-percentile threshold. This figure is not an OOD performance comparison.*

3.2 U-Matrix and Comparison Methods

For unit j with grid neighbors $\mathcal{N}(j)$, a neighborhood-distance summary is

$$U_j = \frac{1}{|\mathcal{N}(j)|} \sum_{k \in \mathcal{N}(j)} \|w_j - w_k\|_2. \tag{2}$$

The example-level U-Matrix signal uses the score of an example's BMU neighborhood. It measures prototype-to-prototype separation, not necessarily distance of the example from its BMU. The original historical five-method SVHN comparison additionally includes Isolation Forest, One-Class SVM and feature autoencoder reconstruction error. Subsequent comparisons add MSP, energy, Mahalanobis and k-nearest-neighbor distance on the frozen CNN representation. Different historical fitting budgets remain documented limitations rather than being silently equalized.

4. Quantitative Evaluation

We report AUROC as a threshold-independent ranking measure. AUPR uses OOD as the positive class and depends on class prevalence. FPR@95TPR denotes the minimum observed ROC false-positive rate at an operating point with TPR at least 0.95; it must not be confused with the

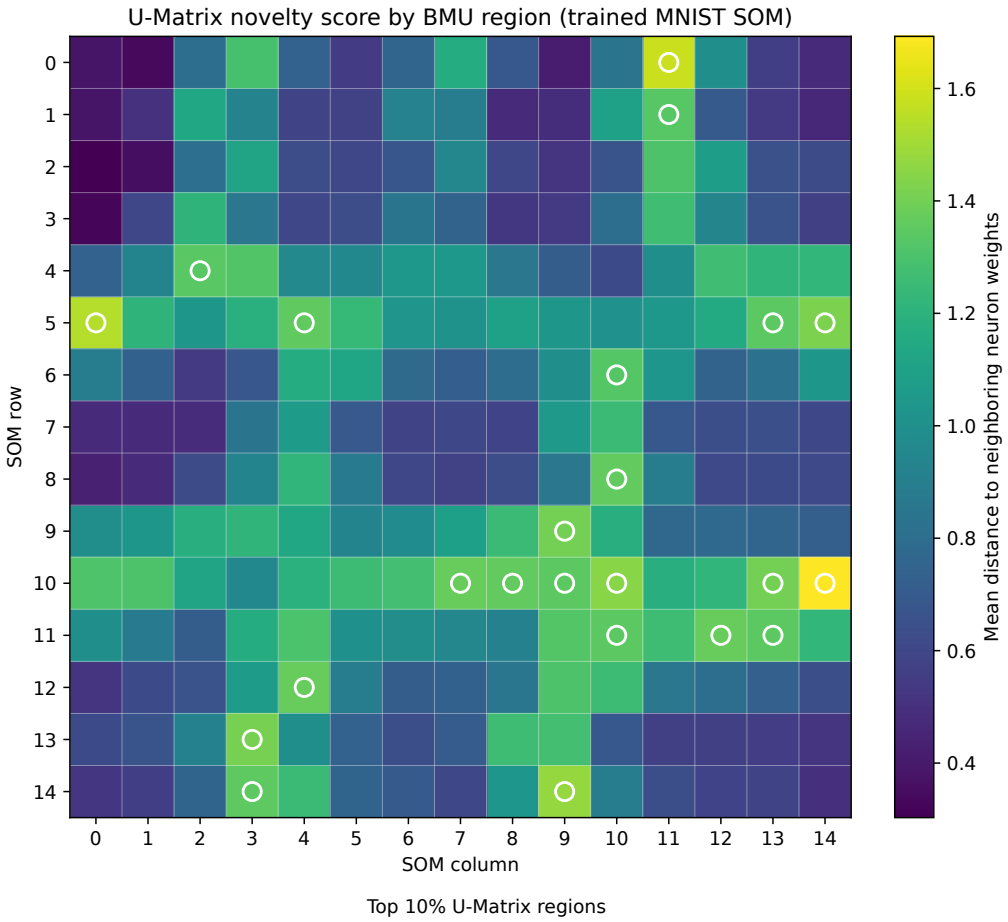


Figure 2. *Recorded U-Matrix overview for the* 15×15 *SOM. Larger local prototype separations do not by themselves prove that individual samples are OOD.*

false-positive rate at the previously frozen QE threshold (Fawcett 2006; Saito and Rehmsmeier 2015). We do not use observed external-OOD results to change historical model settings.

4.1 Primary MNIST versus Fashion-MNIST comparison

The evaluation uses 10,500 MNIST validation negatives and 10,000 Fashion-MNIST test positives. The published point estimates and 95% percentile bootstrap AUROC intervals below are from the recorded frozen-model evaluation with 2,000 stratified bootstrap iterations and seed 2026. The

Table 1. *Fashion-MNIST OOD ranking against MNIST validation. The intervals reflect resampling uncertainty for a fixed representation, not multi-seed training variation.*

Method	AUROC	95% bootstrap CI
Energy	0.995678	[0.995057, 0.996299]
MSP	0.988127	[0.986964, 0.989391]
kNN	0.939197	[0.935631, 0.942673]
Quantization Error	0.918486	[0.914368, 0.922389]
Mahalanobis	0.733506	[0.725800, 0.741031]

measured Fashion-MNIST comparison shows that energy and MSP provide stronger AUROC than QE for the evaluated frozen CNN. Thus QE must not be characterized as the best overall detector. Separate evaluation details and reproducibility evidence are preserved in run-specific small metrics files rather than overwritten historical records.

4.2 Historical SVHN comparison

The original five-method benchmark uses the same 10,500 MNIST validation examples and 26,032 SVHN test examples, with the frozen MNIST representation. The table below records the historical benchmark; it is not retroactively merged into a single equal-budget experiment with the newer methods. At the historical frozen QE threshold, the recorded detector flagged 25,036 of 26,032

Table 2. *Historical SVHN benchmark (one recorded seed). FPR@95TPR values are proportions, not percentages.*

Method	AUROC	AUPR	FPR@95TPR
QE	0.9811	0.9766	0.0453
U-Matrix	0.9696	0.9865	0.0701
Isolation Forest	0.2383	0.5552	0.8718
One-Class SVM	0.9604	0.9764	0.1794
Feature autoencoder	0.0077	0.4979	0.9999

SVHN cases (96.17% detection) and exceeded the threshold for 5.13% of MNIST validation. These two figures describe one fixed threshold, whereas the AUROC and FPR@95TPR results above evaluate ranking or another operating point.

4.3 Additional SVHN baselines

Newer, separately recorded SVHN results provide energy AUROC 0.999797 (AUPR 0.999917), MSP AUROC 0.998495 (AUPR 0.999358), and Mahalanobis AUROC 0.776437 (AUPR 0.785875). These findings again show that energy and MSP outperform QE in this evaluated setting. Because historical model fitting budgets and experimental chronology differ, they should not be presented as a prospectively frozen all-method competition.

5. U-Matrix Association and Failure Analysis

Across 10,500 MNIST validation samples, QE and the U-Matrix BMU score have a recorded Spearman rank correlation of 0.4396. The moderate association is consistent with related but distinct SOM geometry signals. The historical QE experiment missed 996 SVHN samples at its

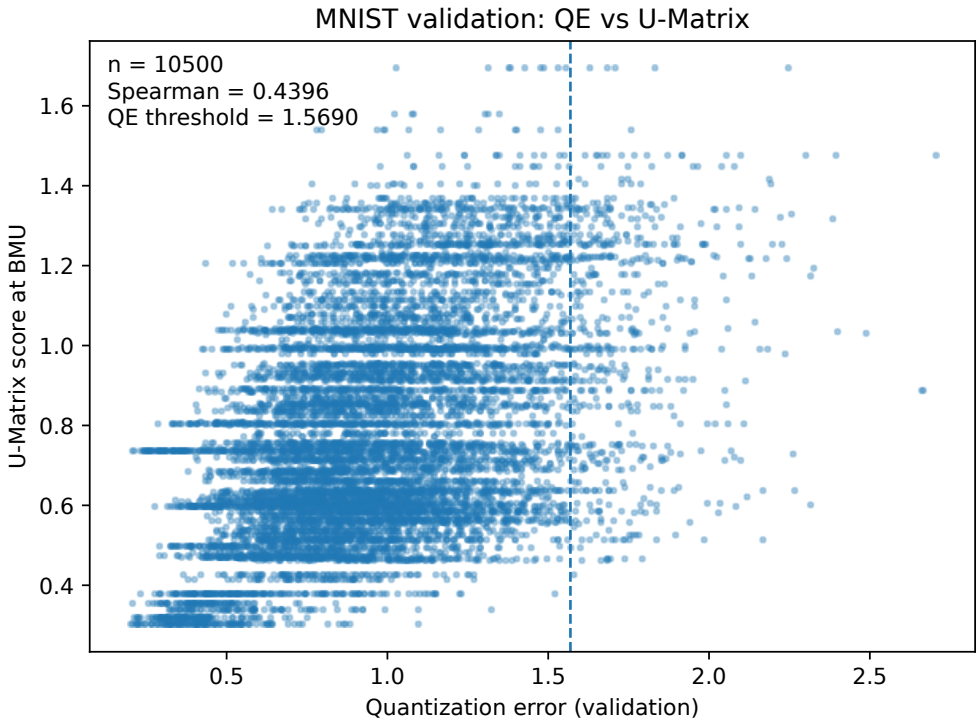


Figure 3. *Recorded QE versus U-Matrix association on MNIST validation; the figure is a development-only comparison.*

frozen threshold. Several low-scoring examples have clean, MNIST-like numeral strokes; this visual pattern is a hypothesis about representation similarity, not a causal explanation. The original Isolation Forest and autoencoder AUROCs fall substantially below 0.5 under their recorded score direction. Their root causes remain unverified; the report retains the original values rather than reversing signs after seeing labels.

6. Leakage Control and Reproducibility

The CNN and SOM were frozen for the documented Phase 3 comparisons. Training embeddings supplied model fitting and the training-derived QE threshold. MNIST validation supplied ID evaluation examples; the official MNIST final-test split was not used for the reported Phase 3 results. External Fashion-MNIST and SVHN test samples supplied observed OOD evaluation evidence, not inputs for post-hoc fitting or threshold selection. As these external sets have now been examined, further configuration changes using their outcomes would be exploratory and require a new protocol and run IDs; they must not be represented as independent validation.

The preregistration document's recorded writing date does not establish from Git history alone that its criteria were fixed before all external OOD results were examined. Historical JSON evidence has been preserved under run-specific directories with a relocation record. The latest EC2 verification completed 74 Phase 3 tests and 172 full MNIST tests successfully, with clean Phase 3 Ruff checks; the earlier Phase 1 data-path errors were resolved by running the suite from the appropriate working directory without changing Phase 1 code. SHA-256 input provenance and run-ID overwrite preflight remain part of the verification procedure.

A separate, newly identified verification run (`mnist_historical_baseline_verification_s42_v1`)



Figure 4. *Recorded lowest-QE SVHN examples missed at the original frozen threshold. These examples are descriptive, not a complete explanation of model behavior.*

reproduced all seven historical MNIST-validation summary measurements with zero absolute numerical difference: Isolation Forest and One-Class SVM score means and standard deviations, and the feature autoencoder initial loss, final loss and mean validation reconstruction error. Frozen training and validation embedding SHA-256 fingerprints matched the references. This supports numerical agreement under the currently committed settings, not byte-identical regeneration of the original historical JSON, checkpoint, or execution environment. The original autoencoder checkpoint and historical package-version manifest remain unavailable; old metrics were not overwritten. No final MNIST test or external OOD samples were accessed during this validation-only reproduction.

7. Limitations and Interpretation

This draft reports fixed-model bootstrap sampling intervals but not variation across at least five independent training seeds. The original One-Class SVM used 10,000 MNIST training embeddings whereas some other fitted baselines used 49,000. Input scale differences for the historical autoencoder and suspicious reversed ranking for two baseline methods require controlled follow-up rather than retrospective correction. No final locked MNIST test-set result is claimed. The comparison demonstrates useful novelty ranking on the examined distributions, not universal or clinically transferable OOD behavior.

8. Conclusion and Next Phase

The principal Fashion-MNIST comparison finds energy and MSP above SOM QE in AUROC, with QE still providing a meaningful prototype-distance novelty signal. The historical SVHN benchmark shows strong QE discrimination relative to its original comparison set, while additional SVHN baselines surpass QE. A future protocol-compliant study should equalize fitting budgets, examine multiple seeds and controls, resolve historical reproduction gaps, and fix the final selection rules before using the untouched MNIST final-test split. These Phase 3 outputs can inform the project's subsequent guided-retraining work without prematurely claiming final benchmark success.

References

Kohonen, T. (1990). The Self-Organizing Map. *Proceedings of the IEEE* 78.9, pp. 1464–1480.

Vesanto, J. and Alhoniemi, E. (2000). Clustering of the Self-Organizing Map. *IEEE Transactions on Neural Networks* 11.3, pp. 586–600.

LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-Based Learning Applied to Document Recognition. *Proceedings of the IEEE* 86.11, pp. 2278–2324.

Jafari, A. (n.d.). NNSOM: Neural Network Self-Organizing Maps in Python. The George Washington University. URL: https://amir-jafari.github.io/SOM/.

Hendrycks, D. and Gimpel, K. (2017). "A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks". *International Conference on Learning Representations*. URL: https://openreview.net/forum?id=Hkg4TI9xl.

Lee, K., Lee, K., Lee, H., and Shin, J. (2018). "A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks". *Advances in Neural Information Processing Systems*. Vol. 31.

Liu, W., Wang, X., Owens, J., and Li, Y. (2020). "Energy-Based Out-of-Distribution Detection". *Advances in Neural Information Processing Systems*. Vol. 33.

Liu, F. T., Ting, K. M., and Zhou, Z.-H. (2008). "Isolation Forest". *2008 Eighth IEEE International Conference on Data Mining*, pp. 413–422.

Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., and Williamson, R. C. (2001). Estimating the Support of a High-Dimensional Distribution. *Neural Computation* 13.7, pp. 1443–1471.

Sakurada, M. and Yairi, T. (2014). "Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction". *Proceedings of the MLSDA 2014 2nd Workshop on Machine Learning for Sensory Data Analysis*, pp. 4–11.

Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). "Out-of-Distribution Detection with Deep Nearest Neighbors". *Proceedings of the 39th International Conference on Machine Learning*. Vol. 162. Proceedings of Machine Learning Research, pp. 20827–20840.

Fawcett, T. (2006). An Introduction to ROC Analysis. *Pattern Recognition Letters* 27.8, pp. 861–874.

Saito, T. and Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE* 10.3, e0118432.